

Streamlining Urban CFD Simulations: An Automated Workflow for SOD2D pre-processing

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² LS-CFD – Large-Scale Computational Fluid Dynamics,
Barcelona Supercomputing Center

OUTLINE

BACKGROUND

- Expansion for Digital Twins

EXPLORATION

- City4CFD
- Open source datasets

METHODOLOGY

- City4CFD
- ANSA scripting

SIMULATION

- CEDVAL case
- Monterrey case
- Terrassa case

FUTURE WORK

- FLIC

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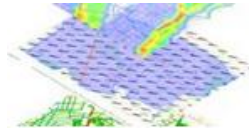
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Urban Digital Twins as the future for urban planning

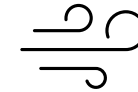
Urban Digital Twins are virtual replicas of cities that integrate data from multiple sources as **Layers of Information** to test and foresee urban plans digitally.



← **Meteorology**

Impact →

- Health impact (Air quality).
- Energy consumption.
- Pollution dispersion.



Limitations

Takes too much time to pre-process.

Lacks guidelines for state-of-the-art solvers as SOD2D.



Barriers to reduce the pollution



Public health strategies

OBJECTIVE

To automatize and generalize pre-processing for urban flow simulations in SOD2D.



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City4CFD & Public Datasets

City4CFD (TUDELf)



Open source



Easy integration
with other
programming
languages

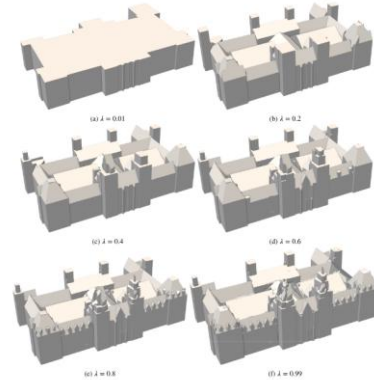
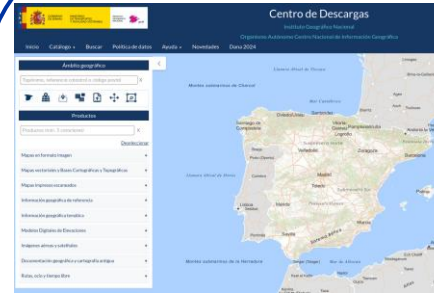


Fig. 5. LiDAR reconstruction with different completeness factors.

Ivan Pađen, Ravi Peters, Clara García-Sánchez, and Hugo Ledoux (2024). Automatic high-detailed building reconstruction workflow for urban microscale simulations. Building and Environment, 265, 2024

Public Datasets



AMB Àrea Metropolitana
de Barcelona



Public access



Long-term plans

LiDAR-PNOA-cob3 2022-2025 CC-BY 4.0 Comunidades autónomas y Instituto Geográfico Nacional y O.A. CNIG, D.G. de Bosques y Desertificación, FEAGA, D.G. del Catastro y ENAIRE
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City4CFD-ANSA Workflow

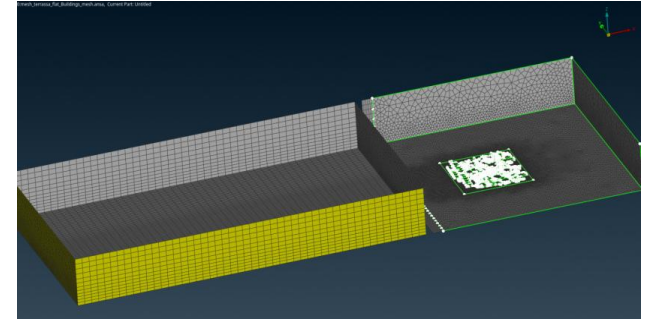
From LiDAR databases to CFD-ready meshes



Generic LiDAR survey points



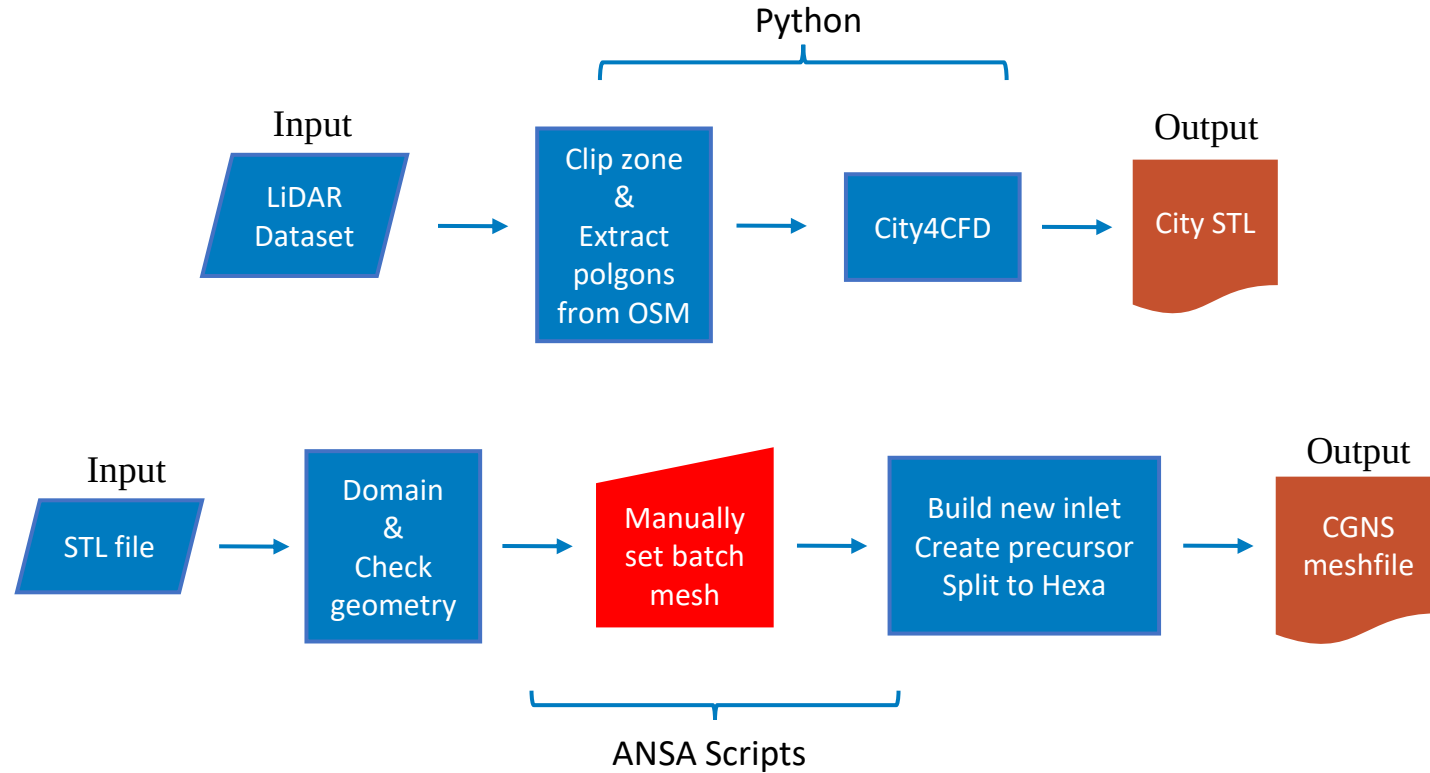
- Python scripts
- City4CFD
- ANSA scripts



SOD2D ready mesh (flat terrain)

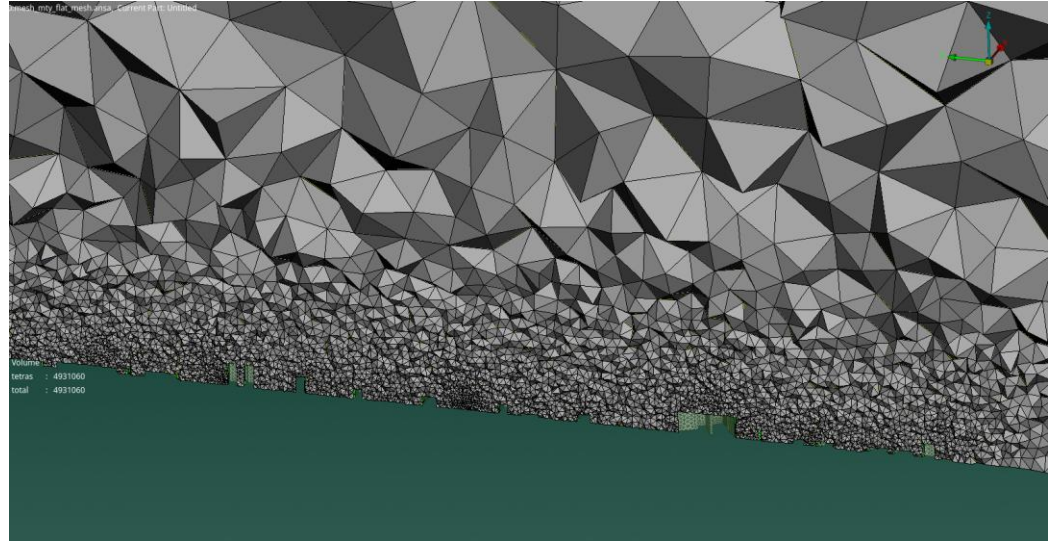


City4CFD-ANSA Workflow



CFD domain characteristics

- SOD2D quality criteria.
- Parametric domain and size boxes based on building height.
- Flat terrain.
- Minutes ready mesh.



Monterrey Volume Mesh

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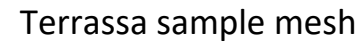
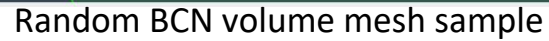
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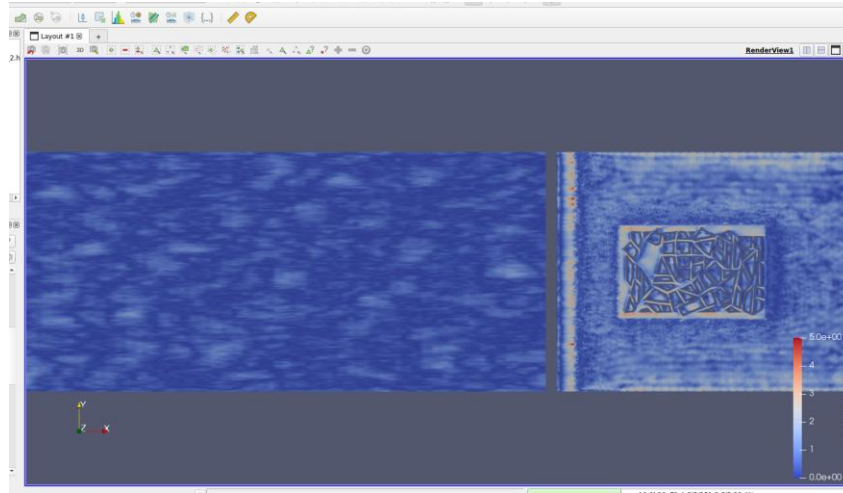
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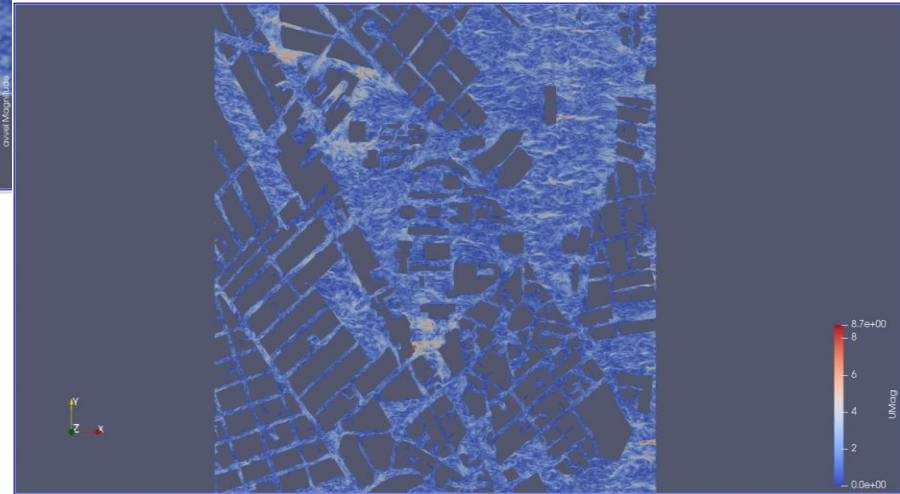


Simulation results examples

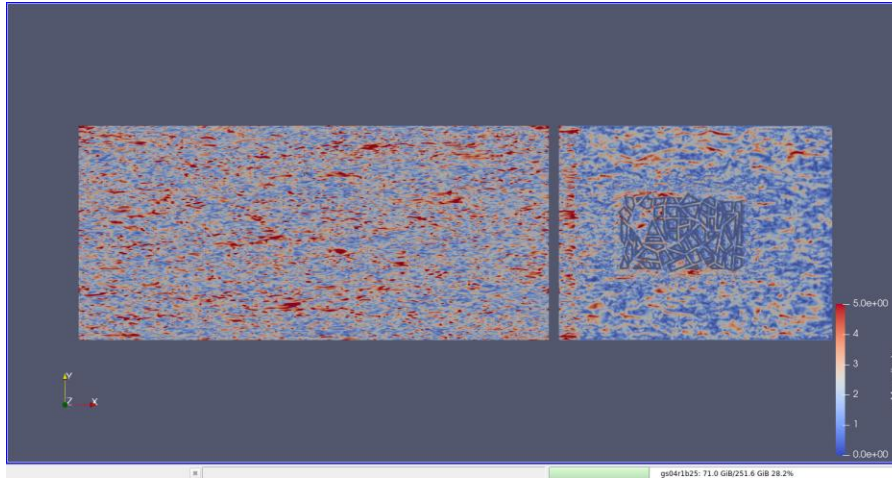


CEDVAL

Monterrey



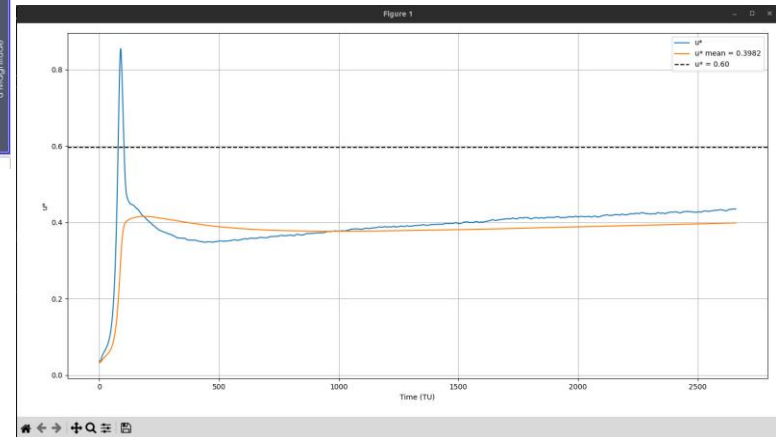
CEDVAL Test Results



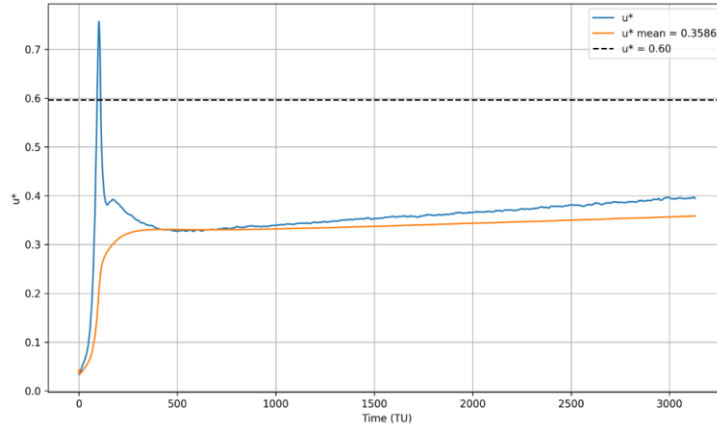
CEDVAL instant

Replica of Cedval simulation with the mesh generated by the workflow.

U^* calculation

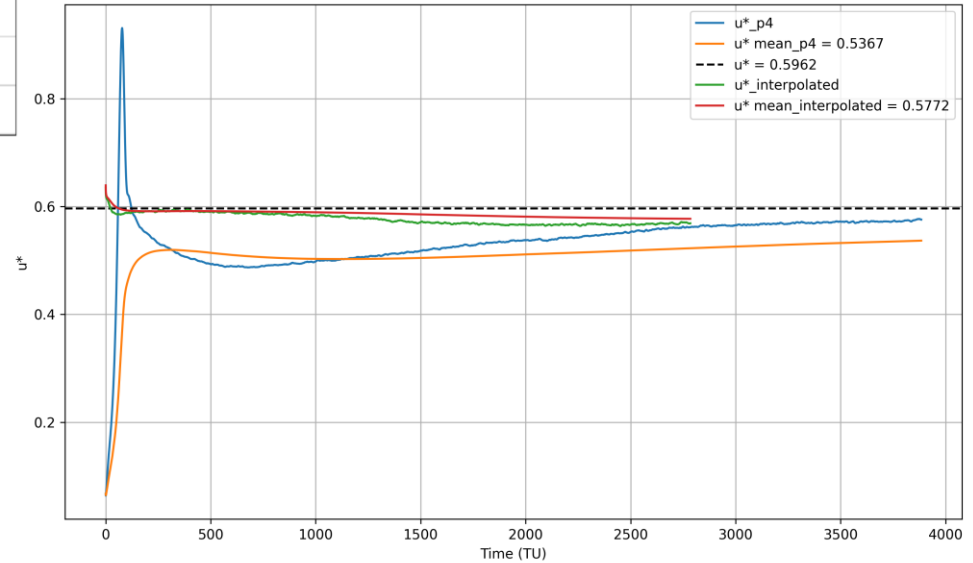


Faster convergence of U^* by interpolating P2 to P4



Order 2 (40 million nodes) simulation for 3000 TU

P2 (40M) interpolated to P4 (250M) vs starting from P4 (250M nodes)



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FLIC — Fast modeLing for Integrated Climate

A modular toolbox for cities climate modeling and prediction

Pre-processing

- City4CFD
- STLtoGEOtool
- ANSA Scripting

Processing

- SOD2D
- Wind Neural Network
- GAVI
- LAMINE
- Etc...



THANK YOU!



TITLE

Autors***

1

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UNIVERSITAT POLITÈCNICA
DE CATALUNYA
BARCELONATECH



Conference Name

Date

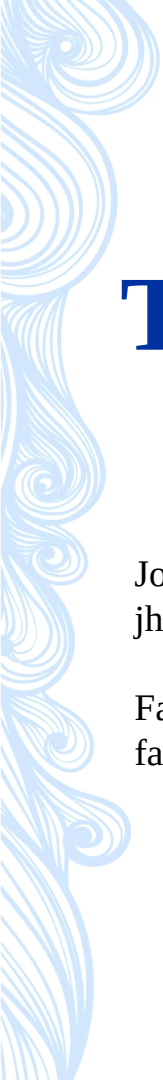


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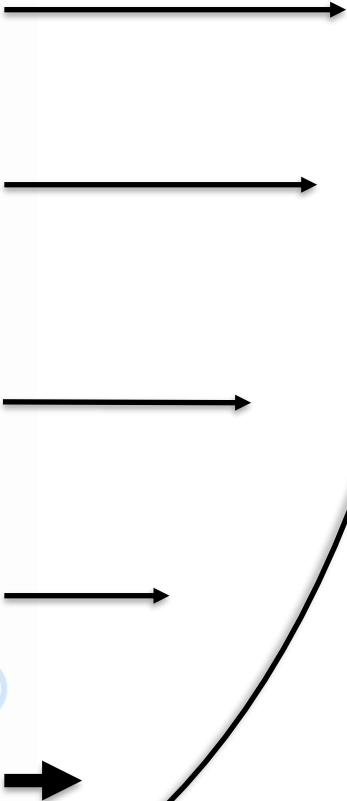
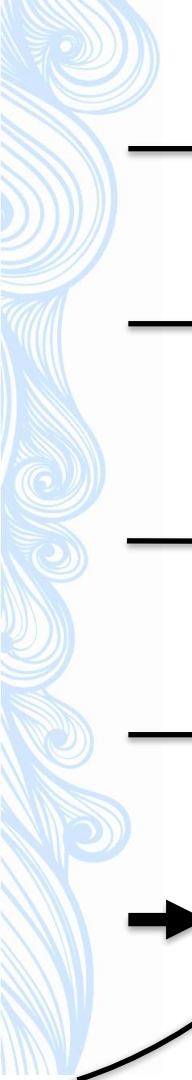




Thank you!

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APPENDIX